

RC Car V1 Sidewalk Autonomy Spec

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This document captures the key design questions, the recorded answers, and the resulting version 1 sidewalk autonomy specification for the RC car project.

1. Project Goal

The goal of version 1 is to drive an RC car autonomously along sidewalks in one overall direction, including through turns, while prioritizing staying on the sidewalk and avoiding sidewalk departure.

Version 1 should use:

- classical vision for sidewalk detection
- LiDAR for obstacle geometry and local heading refinement
- front ultrasonic as emergency stop support
- hall sensor for speed and stuck detection

The system should be modular so ROS2, learned perception, SLAM, or more advanced planning can be added later without rewriting the core control structure.

2. Hardware Baseline

The planned interim hardware set is:

- Raspberry Pi
- Front USB webcam
- LiDAR
- Motor controller
- Drive motors
- Steering servo
- Main battery and power wiring
- Controller
- Hall sensor
- Front ultrasonic sensor

3. Design Questions And Answers

3.1 Path Preference

Question: Do you want the car to follow the center of the sidewalk, or one edge consistently?

Answer: Center of sidewalk.

3.2 Turn Style

Question: If one edge is used during a turn, which edge should it prefer?

Answer: When turning, the car can drive next to an edge the way F1 cars use edges on tracks to make turns smoother.

Interpretation: The center is the default target, but temporary edge bias is allowed during turns if it improves smoothness while remaining on the sidewalk.

3.3 Direction Preference At Turns

Question: At a sidewalk turn, should it always prefer continuing the same overall direction of travel if that path still exists?

Answer: Yes.

3.4 Driveway Cuts

Question: At a driveway cut where the curb disappears, should it keep following the sidewalk path through it?

Answer: No. It should stop and brake until manual control overrides the braking.

3.5 Partial Blockage

Question: If the sidewalk is partially blocked, should it stop or try to go around while staying on the sidewalk?

Answer: Try to go around while on sidewalk.

3.6 Full Blockage

Question: If the sidewalk is fully blocked, should it stop and wait, or attempt a recovery/search behavior?

Answer: Stop and brake until manual control overrides.

3.7 Reverse

Question: Do you want reverse maneuvers allowed at all, or only forward plus stop?

Answer: Any turns can happen.

Interpretation: The system is not limited to a forward-only motion model forever, but version 1 behavior is still primarily sidewalk following plus stopping.

3.8 Initial Autonomous Speed

Question: What is the maximum autonomous speed you want for the first real version?

Answer: The car can go up to 3 mph, but for safety and precision, 2 mph should be the top speed.

3.9 Camera Versus LiDAR

Question: Should LiDAR always override the camera if the camera wants to keep going but LiDAR says the path is unsafe?

Answer summary: The camera sees more than the LiDAR because LiDAR is mounted on top of the robot and is only 2D. The camera is wide and provides richer environmental understanding.

Interpretation: The camera should define sidewalk understanding and intended path. LiDAR should still be used for obstacle avoidance and local geometry, not as the sole path source.

3.10 Shade Handling

Question: When the camera gets confused by shade, should the car slow down and keep using LiDAR guidance, or stop sooner?

Answer: The camera should have experience with shade and should handle it.

Later refined answer: When sidewalk boundaries become visually ambiguous but the surface still appears safe, the RC car should slow down while using its last confident heading as a temporary guide, actively seeking to regain perception confidence before proceeding at normal speed.

3.11 Version 1 Scope

Question: For version 1, do you want obstacle avoidance or just sidewalk following plus stop-on-obstacle?

Answer: Sidewalk-following plus stop-on-obstacle.

3.12 Road Rule

Question: Is the road always forbidden, even if LiDAR says it is open?

Answer: Road is always forbidden.

3.13 Pedestrians

Question: If a pedestrian is on the sidewalk, should the car stop and wait, or pass them if there is enough sidewalk width?

Answer: Pass them if there is enough sidewalk width.

3.14 Manual Override

Question: Do you want autonomous mode to disengage immediately on any manual steering input?

Answer: Autonomy should disengage on any manual steering, throttle, or brake.

3.15 Steering Representation

Question: Should the controller be built around one fused `target_heading` instead of separate steering commands?

Answer: There is only one servo, so all steering has to become one final value at some point.

Interpretation: Yes. The architecture should produce one fused steering target.

3.16 Generalization

Question: Is the first target one specific sidewalk route, or should it work broadly right away?

Answer: It should work on all types of sidewalks.

3.17 Perception Strategy

Question: Are you okay starting with classical vision first, before any trained model?

Answer: Yes for version 1, but the long-term system should be structured like a real modular autonomy stack that can later incorporate ROS2, OpenCV, learned perception, planning, control, and additional middleware.

3.18 First Success Metric

Question: What matters most as the first success metric: handling turns, surviving shade, avoiding sidewalk departure, or not stopping unnecessarily?

Answer: Avoiding sidewalk departure and staying on the drivable path is the most important first success metric.

3.19 Passing Clearance

Question: What minimum clearance should be required to pass an obstacle on the sidewalk?

Answer: The car width is about 200 mm, and the required gap clearance should be 100 mm.

Interpretation: The system should require approximately:

- car width: 200 mm
- side clearance: 100 mm on each side
- total minimum pass width: 400 mm

4. V1 Autonomy Spec

4.1 Core Mission

The RC car should autonomously follow sidewalks while staying on the sidewalk and avoiding sidewalk departure above all else.

4.2 Primary Safety Rule

The highest-priority rule for version 1 is:

Never intentionally select a path that leaves the sidewalk.

This outranks smoothness, speed, or convenience.

4.3 Primary Path Rule

The normal target path should be the center of the sidewalk.

During turns, the controller may temporarily bias closer to one sidewalk edge if that creates a smoother and safer turn while remaining fully on the sidewalk.

4.4 Direction Selection Rule

At turns and sidewalk intersections, the controller should prefer the continuation that best matches the current direction of travel, while also incorporating camera perception of the sidewalk path.

This means:

- continuing forward-ish is preferred over abrupt reorientation
- the camera should influence which forward-ish continuation looks like the real sidewalk

4.5 Driveway Cut Rule

If the sidewalk enters a driveway cut or a region where the curb disappears and the sidewalk boundary becomes invalid in a way that matches the driveway-cut case, the car should:

1. stop
2. brake
3. remain stopped until manual override disengages autonomy

4.6 Partial Obstacle Rule

If an obstacle partially blocks the sidewalk:

- the car may attempt to pass it
- it must remain on the sidewalk
- it must not use the road as a bypass

4.7 Full Obstacle Rule

If the sidewalk is fully blocked:

1. stop
2. brake
3. remain stopped until manual override

No autonomous search or blind departure behavior is allowed for this case in version 1.

4.8 Pedestrian Rule

Pedestrians may be passed only if the remaining sidewalk width supports safe passage.

The required minimum pass width is:

- 200 mm car width
- 100 mm clearance on one side
- 100 mm clearance on the other side

Total required width:

- 400 mm minimum usable passage width

If this clearance is not available, the car must stop and brake.

4.9 Road Rule

The road is always forbidden.

Even if LiDAR reports open space in the road direction, that direction must not be selected as a valid drivable heading.

4.10 Camera Ambiguity Rule

If sidewalk boundaries become visually ambiguous but the surface still appears safe:

1. reduce speed
2. continue temporarily using the last confident heading
3. actively attempt to regain camera confidence
4. resume normal speed only after confidence recovers

This is a slowdown-and-recover behavior, not a full stop by default.

4.11 Fusion Rule

Version 1 should use a fused steering/heading architecture.

Sensor responsibilities:

- Camera: sidewalk understanding, path shape, boundary confidence, sidewalk preference
- LiDAR: obstacle geometry, open-space refinement, collision prevention, local heading refinement
- Front ultrasonic: short-range emergency stop support
- Hall sensor: speed estimation and stuck detection

Fusion philosophy:

- the camera is the primary sidewalk-path sensor
- LiDAR supports safe heading refinement and obstacle checks
- LiDAR should not redefine the road as drivable if the camera says it is not sidewalk
- all steering must resolve into one final fused heading and one final servo command

4.12 Manual Override Rule

Autonomy must disengage immediately on any of the following:

- manual steering input
- manual throttle input
- manual brake input

4.13 Version 1 Scope Rule

Version 1 is defined as:

- sidewalk following
- staying on sidewalk
- stop on driveway cuts
- stop on full blockage
- pass partial obstacles only if safe
- pass pedestrians only if enough width exists

It is not yet defined as full general-purpose autonomous navigation.

4.14 Speed Rule

Initial top autonomous speed for version 1:

- 2 mph maximum

Speed should reduce below that when confidence is reduced or the situation becomes more constrained.

4.15 First Success Metric

The first success metric is:

Avoiding sidewalk departure and staying on the drivable sidewalk path.

Other metrics such as smooth turn handling, shade robustness, and minimizing unnecessary stops are secondary until the baseline safety rule is reliable.

5. Recommended Version 1 Internal Architecture

The recommended version 1 internal architecture is:

- camera sidewalk estimator
- LiDAR obstacle/open-heading evaluator
- heading fusion layer
- speed policy layer
- low-level steering and throttle control

Recommended output representation:

- target_heading
- speed_command
- confidence
- stop_reason if needed

This is better than treating camera and LiDAR as independent raw steering sources.

6. Long-Term Direction

Version 1 should remain simple, but the codebase should stay modular so future versions can add:

- ROS2
- learned perception
- semantic segmentation
- SLAM
- more advanced planning
- improved control methods

The long-term system can grow into a more formal autonomy stack, but version 1 should focus on safe sidewalk following with clear rules and conservative behavior.

7. Summary

The version 1 autonomy design is now defined as:

- classical vision first
- camera-led sidewalk understanding
- LiDAR-supported obstacle geometry
- no road departure
- stop on driveway cuts
- stop on full blockage
- pass only when enough width exists
- immediate manual override
- 2 mph maximum
- first success metric: stay on the sidewalk